

INDIAN STATISTICAL INSTITUTE
Assignment-5 (Mathematics III)
Bachelor of Statistical Data Science (BSDS)

1. Let X be an infinite set and let $\mathcal{T} = \{U \in \mathcal{P}(X) : U = \phi \text{ or } X \setminus U \text{ is finite}\}$. Prove that $\mathcal{T}$ is a topology on X .
2. Let $(X, \mathcal{T})$ be a topological space. Prove that $\mathcal{T}$ is the discrete topology on X if and only if for each $x \in X$, $\{x\} \in \mathcal{T}$.
3. Give an example of a set X and topologies $\mathcal{T}_1$ and $\mathcal{T}_2$ on X such that $\mathcal{T}_1 \cup \mathcal{T}_2$ is not a topology on X .
4. Let (X, d) be a metric space. Define a function $d' : X \times X \rightarrow \mathbb{R}$ by

$$d'(x, y) = \frac{d(x, y)}{1 + d(x, y)}.$$

Prove that (X, d') is a metric space.

5. Prove that the taxicab metric is a metric on $\mathbb{R}^n$.
6. Let (X, d) and (Y, ρ) be metric spaces. An isometry is a one-to-one function f that maps X onto Y so that $d(x, y) = \rho(f(x), f(y))$ for all $x, y \in X$. Prove that if $F : X \rightarrow Y$ is an isometry, then f and f^{-1} are continuous.
7. Let (X, d) be a metric space and let A be a nonempty subset of X . Define $f : X \rightarrow \mathbb{R}$ by $f(x) = \inf\{d(x, a) : a \in A\}$. Prove that f is continuous.
8. Let (X, d_1) and (Y, d_2) be metric spaces, and let $f : X \rightarrow Y$ be a function. Prove that f is continuous if and only if, for each open subset U of (Y, d_2) , $f^{-1}(U)$ is an open subset of (X, d_1) .
9. Let A and B be subsets of a topological space $(X, \mathcal{T})$. Prove that :
 - (a) $\bar{\bar{A}} = \bar{A}$.
 - (b) if $A \subset B$ then $\bar{A} \subset \bar{B}$.
 - (c) $\overline{A \cap B} \subset \bar{A} \cap \bar{B}$.
10. Let $A = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \geq 1\} \cup (0, 0)\}$ and let d be the usual metric on A . Find the closure $B_d((0, 0), 1)$. Justify your answer.
11. Let $\mathcal{T} = \{U \in \mathcal{P}(\mathbb{R}) : 0 \notin U \text{ or } U = \mathbb{R}\}$.
 - (a) Prove that $\mathcal{T}$ is a topology on $\mathbb{R}$.
 - (b) Describe the closed subsets of $(\mathbb{R}, \mathcal{T})$.
 - (c) Find $\{\bar{1}\}$.